

# Appendix 5 - Equipment List

|                                                                                     |                        | Revision                                          | Date      | Description              | By | Checked                                                                                                                                                                                                                                                                                                                                               | Approved | DOCUMENT NUMBER   | REV |
|-------------------------------------------------------------------------------------|------------------------|---------------------------------------------------|-----------|--------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------|-----|
| Client                                                                              | Asymchem               |                                                   |           |                          |    |                                                                                                                                                                                                                                                                                                                                                       |          | 300291-PR-SC-0001 | A1  |
| Location                                                                            | Sandwich, UK           |                                                   |           |                          |    |                                                                                                                                                                                                                                                                                                                                                       |          |                   |     |
| Project Name                                                                        | PDF Extension Facility |                                                   |           |                          |    |                                                                                                                                                                                                                                                                                                                                                       |          |                   |     |
| Project No.                                                                         | 300291                 | A1                                                | 21 May 26 | Feasibility Design Issue | ME | MC                                                                                                                                                                                                                                                                                                                                                    | ME       |                   |     |
| <div>ASYMCHEM PDF EXTENSION FACILITY</div> <div>EQUIPMENT LIST</div>                |                        |                                                   |           |                          |    |                                                                                                                                                                                                                                                                                                                                                       |          |                   |     |
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| EQUIPMENT LIST                  |        |         |                         |                                                                                                                                                   |      |                                                              |                                    |                                                                |           |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           |                   |     |    |
|---------------------------------|--------|---------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------|------------------------------------|----------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----|----|
|                                 |        |         |                         | Revision                                                                                                                                          | Date | Description                                                  | By                                 | Checked                                                        | Approved  |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | DOCUMENT NUMBER   | REV |    |
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| Location                        |        |         |                         | Sandwich, UK                                                                                                                                      |      |                                                              |                                    |                                                                |           |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           |                   |     |    |
| Project Name                    |        |         |                         | PDF Extension Facility                                                                                                                            |      |                                                              |                                    |                                                                |           |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           |                   |     |    |
| Project No.                     |        |         |                         | 300291                                                                                                                                            |      | A1                                                           | 21 May 26                          | Feasibility Design Issue                                       | ME        |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           |                   |     |    |
| Tag                             | ID No. | No. Off | DESCRIPTION             | TECHNICAL CHARACTERISTICS - SIZE/DUTY                                                                                                             |      | MATERIALS OF CONSTRUCTION                                    | WEIGHT (kg)                        | DIMENSIONS (mm) W x D x H                                      | LOCATION  | Process                                                                                                                                                                                                                                                                                                                                               | Electrical Load     | NOTES                                                                                                                                     | REVISION          |     |    |
| REACTORS & ASSOCIATED EQUIPMENT |        |         |                         |                                                                                                                                                   |      |                                                              |                                    |                                                                |           |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           |                   |     |    |
| R25                             |        | 1       | Reactor                 | 5000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |      | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 | Empty - 6320<br>Full water - 12170 | Internal<br>1750 D x 2525 H<br><br>External<br>1920 D x 5228 H | 2nd floor | Y - Transfer from<br>reagent, slurries and<br>liquids & solvents from<br>other process areas                                                                                                                                                                                                                                                          | 7.5 kW electromotor |                                                                                                                                           | A1                |     |    |
| R25                             | PC01   | 1       | Primary Condenser       | 30m <sup>3</sup>                                                                                                                                  |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     | Review use of tantalum as condenser MOC                                                                                                   | A1                |     |    |
| R25                             | VC01   | 1       | Vent Condenser          | 3m <sup>3</sup>                                                                                                                                   |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     | Review use of tantalum as condenser MOC                                                                                                   | A1                |     |    |
| R25                             | JP01   | 1       | Jacket Circulation Pump |                                                                                                                                                   |      | SS316L                                                       |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R26                             |        | 1       | Reactor                 | 8000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |      | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 | Empty - 8765<br>Full water - 17965 | Internal<br>2000 D x 3025 H<br><br>External<br>2224 D x 6020 H | 2nd floor | Y - Transfer from<br>reagent, slurries and<br>liquids & solvents from<br>other process areas                                                                                                                                                                                                                                                          | 11 kW electromotor  |                                                                                                                                           | A1                |     |    |
| R26                             | PC01   | 1       | Primary Condenser       | 40m <sup>3</sup>                                                                                                                                  |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     | Review use of tantalum as condenser MOC                                                                                                   | A1                |     |    |
| R26                             | VC01   | 1       | Vent Condenser          | 5m <sup>3</sup>                                                                                                                                   |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     | Review use of tantalum as condenser MOC                                                                                                   | A1                |     |    |
| R26                             | JP01   | 1       | Jacket Circulation Pump |                                                                                                                                                   |      | SS316L                                                       |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R27                             |        | 1       | Reactor                 | 5000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |      | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 | Empty - 6320<br>Full water - 12170 | Internal<br>1750 D x 2525 H<br><br>External<br>1920 D x 5228 H | 2nd floor | Y - Transfer from<br>reagent, slurries and<br>liquids & solvents from<br>other process areas                                                                                                                                                                                                                                                          | 7.5 kW electromotor |                                                                                                                                           | A1                |     |    |
| R27                             | PC01   | 1       | Primary Condenser       | 30m <sup>3</sup>                                                                                                                                  |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R27                             | VC01   | 1       | Vent Condenser          | 3m <sup>3</sup>                                                                                                                                   |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R27                             | JP01   | 1       | Jacket Circulation Pump |                                                                                                                                                   |      | SS316L                                                       |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R28                             |        | 1       | Reactor                 | 8000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |      | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 | Empty - 8765<br>Full water - 17965 | Internal<br>2000 D x 3025 H<br><br>External<br>2224 D x 6020 H | 2nd floor | Y - Transfer from<br>reagent, slurries and<br>liquids & solvents from<br>other process areas                                                                                                                                                                                                                                                          | 11 kW electromotor  |                                                                                                                                           | A1                |     |    |
| R28                             | PC01   | 1       | Primary Condenser       | 40m <sup>3</sup>                                                                                                                                  |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R28                             | VC01   | 1       | Vent Condenser          | 5m <sup>3</sup>                                                                                                                                   |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R28                             | JP01   | 1       | Jacket Circulation Pump |                                                                                                                                                   |      | SS316L                                                       |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R29                             |        | 1       | Reactor                 | 2500L<br>Design Temp: -85/165 °C<br>Oper. Temp: -70/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.57 MPa                                    |      | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 |                                    | Internal<br>1600 D x 2290 H<br><br>External<br>1750 D x 4506 H | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       | 5.5 kW electromotor | Review option to have this also act as a cryogenic reactor with hydrogen venting capabilities. Cryo reactors should be in close proximity | A1                |     |    |
| R29                             | PC01   | 1       | Primary Condenser       | 19m <sup>3</sup>                                                                                                                                  |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R29                             | VC01   | 1       | Vent Condenser          | 1.5m <sup>3</sup>                                                                                                                                 |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R29                             | JP01   | 1       | Jacket Circulation Pump | 36m³/h-20m                                                                                                                                        |      | SS316L                                                       |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R29                             | CC01   | 1       | Jacket Cryogenic Cooler |                                                                                                                                                   |      | SS304                                                        |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R30                             |        | 1       | Reactor                 | 2500L working volume (3800L gross volume)<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa |      | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 | Empty - 3972<br>Full water - 7772  | Internal<br>1600 D x 2290 H<br><br>External<br>1750 D x 4506 H | 2nd floor | Y - Transfer from<br>reagent, slurries and<br>liquids & solvents from<br>other process areas                                                                                                                                                                                                                                                          | 5.5 kW electromotor |                                                                                                                                           | A1                |     |    |
| R30                             | PC01   | 1       | Primary Condenser       | 19m <sup>3</sup>                                                                                                                                  |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R30                             | VC01   | 1       | Vent Condenser          | 1.5m <sup>3</sup>                                                                                                                                 |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R30                             | JP01   | 1       | Jacket Circulation Pump | 36m³/h-20m                                                                                                                                        |      | SS304                                                        |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R31                             |        | 1       | Reactor                 | 5000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |      | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 | Empty - 6320<br>Full water - 12170 | Internal<br>1750 D x 2525 H<br><br>External<br>1920 D x 5228 H | 2nd floor | Y - Transfer from<br>reagent, slurries and<br>liquids & solvents from<br>other process areas                                                                                                                                                                                                                                                          | 7.5 kW electromotor |                                                                                                                                           | A1                |     |    |
| R31                             | PC01   | 1       | Primary Condenser       | 30m <sup>3</sup>                                                                                                                                  |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R31                             | VC01   | 1       | Vent Condenser          | 3m <sup>3</sup>                                                                                                                                   |      | Hastelloy C-22                                               |                                    |                                                                | 2nd floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |
| R31                             | JP01   | 1       | Jacket Circulation Pump |                                                                                                                                                   |      | SS304                                                        |                                    |                                                                | 1st floor |                                                                                                                                                                                                                                                                                                                                                       |                     |                                                                                                                                           | A1                |     |    |



EQUIPMENT LIST

| Client                     |        |         | Revision                 | Date                                                                                                                                              | Description | By                                                           | Checked                            | Approved                                                                               | This document contains proprietary information belonging to Scitech and / or affiliated companies and shall be used only for the purpose for which it was supplied. It shall not be copied, reproduced or otherwise used, nor shall such information be supplied to others except with the prior consent of Scitech and shall be returned on request. |                                                                                              |                     | DOCUMENT NUMBER                                               | REV      |
|----------------------------|--------|---------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------|----------|
| Location                   |        |         | Sandwich, UK             |                                                                                                                                                   |             |                                                              |                                    |                                                                                        |                                                                                                                                                                                                                                                                                                                                                       |                                                                                              |                     | 300291-PR-SC-0001                                             | A1       |
| Project Name               |        |         | PDF Extension Facility   |                                                                                                                                                   |             |                                                              |                                    |                                                                                        |                                                                                                                                                                                                                                                                                                                                                       |                                                                                              |                     |                                                               |          |
| Project No.                |        |         | 300291                   |                                                                                                                                                   |             | A1                                                           | 21 May 26                          | Feasibility Design Issue                                                               |                                                                                                                                                                                                                                                                                                                                                       |                                                                                              |                     |                                                               |          |
| Tag                        | ID No. | No. Off | DESCRIPTION              | TECHNICAL CHARACTERISTICS - SIZE/DUTY                                                                                                             |             | MATERIALS OF CONSTRUCTION                                    | WEIGHT (kg)                        | DIMENSIONS (mm) W x D x H                                                              | LOCATION                                                                                                                                                                                                                                                                                                                                              | Process                                                                                      | Electrical Load     | NOTES                                                         | REVISION |
| R32                        |        | 1       | Reactor                  | 5000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/130 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Hastelloy C-22                                               |                                    | Internal<br>1750 D x 2525 H<br><br>External<br>1920 D x 5228 H                         | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | 7.5 kW electromotor |                                                               | A1       |
| R32                        | PC01   | 1       | Primary Condenser        | 30m <sup>2</sup>                                                                                                                                  |             | Hastelloy C-22                                               |                                    |                                                                                        | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R32                        | VC01   | 1       | Vent Condenser           | 3m <sup>2</sup>                                                                                                                                   |             | Hastelloy C-22                                               |                                    |                                                                                        | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R32                        | JP01   | 1       | Jacket Circulation Pump  |                                                                                                                                                   |             | SS304                                                        |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R33                        |        | 1       | Reactor                  | 8000L<br>Design Temp: -85/165 °C<br>Oper. Temp: -70/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Hastelloy C-22                                               |                                    | Internal<br>2000 D x 3025 H<br><br>External<br>2224 D x 6020 H                         | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | 11 kW electromotor  | Review option to have this capable of venting Hydrogen IIC T4 | A1       |
| R33                        | PC01   | 1       | Primary Condenser        | 40m <sup>2</sup>                                                                                                                                  |             | Hastelloy C-22                                               |                                    |                                                                                        | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R33                        | VC01   | 1       | Vent Condenser           | 5m <sup>2</sup>                                                                                                                                   |             | Hastelloy C-22                                               |                                    |                                                                                        | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R33                        | JP01   | 1       | Jacket Circulation Pump  |                                                                                                                                                   |             | SS304                                                        |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R33                        | CC01   | 1       | Jacket Cryogenic Cooler  |                                                                                                                                                   |             | SS304                                                        |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R34                        |        | 1       | Reactor                  | 8000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Glass Lined<br>Vessel and jacket from<br>ASME SA516 Grade 60 | Empty - 8765<br>Full water - 17965 | Internal<br>2000 D x 3025 H<br><br>External<br>2224 D (100+10mm<br>thickness) x 6020 H | 2nd floor                                                                                                                                                                                                                                                                                                                                             | Y - Transfer from<br>reagent, slurries and<br>liquids & solvents from<br>other process areas | 11 kW electromotor  |                                                               | A1       |
| R34                        | PC01   | 1       | Primary Condenser        | 40m <sup>2</sup>                                                                                                                                  |             | Hastelloy C-22                                               |                                    |                                                                                        | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R34                        | VC01   | 1       | Vent Condenser           | 5m <sup>2</sup>                                                                                                                                   |             | Hastelloy C-22                                               |                                    |                                                                                        | 2nd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R34                        | JP01   | 1       | Jacket Circulation Pump  |                                                                                                                                                   |             | SS304                                                        |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| HYDROGENATION              |        |         |                          |                                                                                                                                                   |             |                                                              |                                    |                                                                                        |                                                                                                                                                                                                                                                                                                                                                       |                                                                                              |                     |                                                               |          |
| R35                        |        | 1       | Hydrogenation Reactor    | 2500L working volume (3400L gross volume)<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~2.4 MPa<br>Oper. P: -0.1~2.0 MPa |             | Hastelloy C-22                                               |                                    | Internal<br>1400 D x 2075 H<br><br>External<br>1570 D x 4663 H                         | 3rd floor                                                                                                                                                                                                                                                                                                                                             | Y - Methanol<br>Acetone<br>LIBC?                                                             |                     | Hydrogen venting vessel IIC T4                                | A1       |
| R35                        | VC01   | 1       | Vent Condenser           | 10m <sup>2</sup>                                                                                                                                  |             | Hastelloy C-22                                               |                                    |                                                                                        | 3rd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     | Hydrogen venting vessel IIC T4                                | A1       |
| R35                        | TCU01  | 1       | TCU                      | Welded plate, 92kW                                                                                                                                |             | SS304                                                        |                                    |                                                                                        | 3rd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R35                        | CU1    | 1       | Hydrogenator Cooler Unit | Welded plate, 210 kW                                                                                                                              |             | SS316L                                                       |                                    |                                                                                        | 3rd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| R35                        | JP01   | 1       | Jacket Circulation Pump  |                                                                                                                                                   |             | SS304                                                        |                                    |                                                                                        | 3rd floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | Y                   |                                                               | A1       |
| RECEIVERS & TRANSFER PUMPS |        |         |                          |                                                                                                                                                   |             |                                                              |                                    |                                                                                        |                                                                                                                                                                                                                                                                                                                                                       |                                                                                              |                     |                                                               |          |
| RC01                       |        | 1       | Distillate Receiver      | 1000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Glass Lined                                                  |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| RC01                       | TP01   | 1       | Transfer Pump            | 5.5m <sup>3</sup> /h-20m                                                                                                                          |             | PTFE                                                         |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | Y                   |                                                               | A1       |
| RC02                       |        | 1       | Distillate Receiver      | 1000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Glass Lined                                                  |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| RC02                       | TP01   | 1       | Transfer Pump            | 5.5m <sup>3</sup> /h-20m                                                                                                                          |             | PTFE                                                         |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | Y                   |                                                               | A1       |
| RC03                       |        | 1       | Distillate Receiver      | 1000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Glass Lined                                                  |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| RC03                       | TP01   | 1       | Transfer Pump            | 5.5m <sup>3</sup> /h-20m                                                                                                                          |             | PTFE                                                         |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | Y                   |                                                               | A1       |
| RC04                       |        | 1       | Distillate Receiver      | 1000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Glass Lined                                                  |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| RC04                       | TP01   | 1       | Transfer Pump            | 5.5m <sup>3</sup> /h-20m                                                                                                                          |             | PTFE                                                         |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | Y                   |                                                               | A1       |
| RC05                       |        | 1       | Distillate Receiver      | 1000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                     |             | Glass Lined                                                  |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                     |                                                               | A1       |
| RC05                       | TP01   | 1       | Transfer Pump            | 5.5m <sup>3</sup> /h-20m                                                                                                                          |             | PTFE                                                         |                                    |                                                                                        | 1st floor                                                                                                                                                                                                                                                                                                                                             |                                                                                              | Y                   |                                                               | A1       |



EQUIPMENT LIST

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|-------------------|--------|---------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------|-----------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Location          |        |         | Sandwich, UK                       |                                                                                                                                                                                                                                                                                                                                         |             |                                          |                       |                                    |                                                                                                                                                                                                                                                                                                                                                       |         |                                                            | 300291-PR-SC-0001                                                                                                                                 | A1       |
| Project Name      |        |         | PDF Extension Facility             |                                                                                                                                                                                                                                                                                                                                         |             |                                          |                       |                                    |                                                                                                                                                                                                                                                                                                                                                       |         |                                                            |                                                                                                                                                   |          |
| Project No.       |        |         | 300291                             |                                                                                                                                                                                                                                                                                                                                         |             | A1                                       | 21 May 26             | Feasibility Design Issue           |                                                                                                                                                                                                                                                                                                                                                       |         |                                                            |                                                                                                                                                   |          |
| Tag               | ID No. | No. Off | DESCRIPTION                        | TECHNICAL CHARACTERISTICS - SIZE/DUTY                                                                                                                                                                                                                                                                                                   |             | MATERIALS OF CONSTRUCTION                | WEIGHT (kg)           | DIMENSIONS (mm) W x D x H          | LOCATION                                                                                                                                                                                                                                                                                                                                              | Process | Electrical Load                                            | NOTES                                                                                                                                             | REVISION |
| RC06              |        | 1       | Distillate Receiver                | 1000L<br>Design Temp: -29/200 °C<br>Oper. Temp: -20/150 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                                                                                                                                                                                                           |             | Glass Lined                              |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         |                                                            |                                                                                                                                                   | A1       |
| RC06              | TP01   | 1       | Transfer Pump                      | 5.5m³/h-20m                                                                                                                                                                                                                                                                                                                             |             | PTFE                                     |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         | Y                                                          |                                                                                                                                                   | A1       |
| FILTER / DRYERS   |        |         |                                    |                                                                                                                                                                                                                                                                                                                                         |             |                                          |                       |                                    |                                                                                                                                                                                                                                                                                                                                                       |         |                                                            |                                                                                                                                                   |          |
| FD01              |        | 1       | Filter/Dryer                       | DN1200 (1.13m³)<br>Design Temp: -20/150 °C<br>Oper. Temp: -15/130 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                                                                                                                                                                                                 |             | Hastelloy C-22                           |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD01              | RC01   | 1       | Filtrate Receiver                  | 1000L                                                                                                                                                                                                                                                                                                                                   |             | Glass Lined                              |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD01              | TP01   | 1       | Transfer Pump                      | 5.5m³/h-20m                                                                                                                                                                                                                                                                                                                             |             | PTFE                                     |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD01              | VC01   | 1       | Vent Condenser                     | 5m³                                                                                                                                                                                                                                                                                                                                     |             | Hastelloy C-22                           |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD01              | TCU01  | 1       | TCU                                |                                                                                                                                                                                                                                                                                                                                         |             | SS304                                    |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD02              |        | 1       | Filter/Dryer                       | DN1400 (1.53m³)<br>Design Temp: -20/150 °C<br>Oper. Temp: -15/130 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                                                                                                                                                                                                 |             | Hastelloy C-22                           |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD02              | RC01   | 1       | Filtrate Receiver                  | 1000L                                                                                                                                                                                                                                                                                                                                   |             | Glass Lined                              |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD02              | TP01   | 1       | Transfer Pump                      | 5.5m³/h-20m                                                                                                                                                                                                                                                                                                                             |             | PTFE                                     |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD02              | VC01   | 1       | Vent Condenser                     | 7m³                                                                                                                                                                                                                                                                                                                                     |             | Hastelloy C-22                           |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD02              | TCU01  | 1       | TCU                                |                                                                                                                                                                                                                                                                                                                                         |             | SS304                                    |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD03              |        | 1       | Filter/Dryer                       | DN1400 (1.53m³)<br>Design Temp: -20/150 °C<br>Oper. Temp: -15/130 °C<br>Design P: -0.1~0.6 MPa<br>Oper. P: -0.1~0.3 MPa                                                                                                                                                                                                                 |             | Hastelloy C-22                           |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD03              | RC01   | 1       | Filtrate Receiver                  | 1000L                                                                                                                                                                                                                                                                                                                                   |             | Glass Lined                              |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD03              | TP01   | 1       | Transfer Pump                      | 5.5m³/h-20m                                                                                                                                                                                                                                                                                                                             |             | PTFE                                     |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD03              | VC01   | 1       | Vent Condenser                     | 7m³                                                                                                                                                                                                                                                                                                                                     |             | Hastelloy C-22                           |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| FD03              | TCU01  | 1       | TCU                                |                                                                                                                                                                                                                                                                                                                                         |             | SS304                                    |                       |                                    | Ground floor                                                                                                                                                                                                                                                                                                                                          |         |                                                            |                                                                                                                                                   | A1       |
| VACUUM PUMPS      |        |         |                                    |                                                                                                                                                                                                                                                                                                                                         |             |                                          |                       |                                    |                                                                                                                                                                                                                                                                                                                                                       |         |                                                            |                                                                                                                                                   |          |
| -                 | VP01   | 1       | Vacuum Pump                        | pumping speed : 300L/S<br>limiting vacuum : 1Pa                                                                                                                                                                                                                                                                                         |             | ---                                      |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         |                                                            |                                                                                                                                                   | A1       |
| -                 | VP02   | 1       | Vacuum Pump                        | pumping speed : 300L/S<br>limiting vacuum : 1Pa                                                                                                                                                                                                                                                                                         |             | ---                                      |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         |                                                            |                                                                                                                                                   | A1       |
| -                 | VP03   | 1       | Vacuum Pump                        | pumping speed : 300L/S<br>limiting vacuum : 1Pa                                                                                                                                                                                                                                                                                         |             | ---                                      |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         |                                                            |                                                                                                                                                   | A1       |
| -                 | VP04   | 1       | Vacuum Pump                        | pumping speed : 300L/S<br>limiting vacuum : 1Pa                                                                                                                                                                                                                                                                                         |             | ---                                      |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         |                                                            |                                                                                                                                                   | A1       |
| -                 | VP05   | 1       | Vacuum Pump                        | pumping speed : 300L/S<br>limiting vacuum : 1Pa                                                                                                                                                                                                                                                                                         |             | ---                                      |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         |                                                            |                                                                                                                                                   | A1       |
| -                 | VP06   | 1       | Vacuum Pump                        | pumping speed : 300L/S<br>limiting vacuum : 1Pa                                                                                                                                                                                                                                                                                         |             | ---                                      |                       |                                    | 1st floor                                                                                                                                                                                                                                                                                                                                             |         |                                                            |                                                                                                                                                   | A1       |
| SCRUBBER PACKAGES |        |         |                                    |                                                                                                                                                                                                                                                                                                                                         |             |                                          |                       |                                    |                                                                                                                                                                                                                                                                                                                                                       |         |                                                            |                                                                                                                                                   |          |
|                   |        |         | ACID & Alkali IBC Package          | Run & standby NaOH & Sulfuric acid storage & pumping system. 4 x 1m3 IBC's , 2 bunds with weather covers , dip tubes, distribution pumps, hoses and piping. Controls to interface with scrubber package PLC's, telemetry to central SCADA contol on level and status                                                                    |             | PP/ GRP/ PE/ 316, Painted steel          | 4500 kg TBC           | Nominal 5m W x 1.8m D x 3.0m H TBC | TBC                                                                                                                                                                                                                                                                                                                                                   |         | 1 Ph nominal, 1000 W for lights and possible trace heating |                                                                                                                                                   | A1       |
| SC05              |        | 1       | Alkaline Gas Scrubber Package      | 425 m3/h Air flow (variable). Pressure rating +-55mBar @-10 to =40 C. Scrubber performance Inlet 2%/w max as NH3 Outlet concentration <10mg/m3. Power & Control: Local PLC & MCC with telemetry to central SCADA for status and alarms. Externally Bunded (by others) Include to raised gavanised access platform / skid. ATEX Zone 2 . |             | PP/GRP/HAST/316SS/PTFE- CS               | 5000 kg (Flooded) TBC | Nominal 5m W x 2.5m D x 7.0m H TBC | Externally located                                                                                                                                                                                                                                                                                                                                    |         | Approx 12kW                                                | Acid scrubbing fluid, for alkaline fumes such as NH3. Package shall be cabable of operating as an acid scrubber using an aklaline scrubbing fluid | A1       |
|                   | CU1    | 1       | Alkaline Gas Scrubber Cooler       | Heat exchanger for scrubbing fluid Primary target exit temp <30 C                                                                                                                                                                                                                                                                       |             | Hastelloy C-22 process contact & 304/316 | SC05 Package          | SC05 Package                       | Externally located                                                                                                                                                                                                                                                                                                                                    |         |                                                            |                                                                                                                                                   | A1       |
|                   | FN1    | 1       | Alkaline Gas Scrubber Fan          | 0 to 425m3/h High pressure centrifugal fan design, External Dp 8kPa.                                                                                                                                                                                                                                                                    |             | PP / GRP - (TBC)                         |                       |                                    | Externally located                                                                                                                                                                                                                                                                                                                                    |         |                                                            |                                                                                                                                                   | A1       |
|                   | FN2    | 1       | Stand-by Alkaline Gas Scrubber Fan | 0 to 425m3/h High pressure centrifugal fan design, External Dp 8kPa.                                                                                                                                                                                                                                                                    |             | PP / GRP - (TBC)                         |                       |                                    | Externally located                                                                                                                                                                                                                                                                                                                                    |         |                                                            |                                                                                                                                                   | A1       |
|                   | P1     | 1       | Alkaline Gas Scrubber Pump         | Centrifugal , sealless mag drive                                                                                                                                                                                                                                                                                                        |             | Glass Filled Furane Resin                |                       |                                    | Externally located                                                                                                                                                                                                                                                                                                                                    |         |                                                            |                                                                                                                                                   | A1       |
|                   | P2     | 1       | Alklaine Gas Scrubber Pump         | Centrifugal , sealless mag drive                                                                                                                                                                                                                                                                                                        |             | Glass Filled Furane Resin                |                       |                                    | Externally located                                                                                                                                                                                                                                                                                                                                    |         |                                                            |                                                                                                                                                   | A1       |
|                   | PS1    | 1       | Alkaline Gas Process Scrubber      | Packed bed with demister and cleaning sprays, fluid distribution nozzles and piping to BS EN 13121-3                                                                                                                                                                                                                                    |             | PP / GRP -                               |                       |                                    | Externally located                                                                                                                                                                                                                                                                                                                                    |         |                                                            |                                                                                                                                                   | A1       |
|                   | ST1    | 1       | Alkaline Gas Scrubber Storage Tank | Srubbing fluid vessel with appriate nozzles, suppports and manway to BS EN 13121-3                                                                                                                                                                                                                                                      |             | PP / GRP -                               |                       |                                    | Externally located                                                                                                                                                                                                                                                                                                                                    |         |                                                            |                                                                                                                                                   | A1       |

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|                                                                                                                      |        |         |                                |                                       |                           |             |                            |                    |                                                                                                                                                                                                                                                                                                                                                       |                 |                     |          |
|----------------------------------------------------------------------------------------------------------------------|--------|---------|--------------------------------|---------------------------------------|---------------------------|-------------|----------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|----------|
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|                                                                                                                      |        |         | Revision                       | Date                                  | Description               | By          | Checked                    | Approved           |                                                                                                                                                                                                                                                                                                                                                       |                 |                     |          |
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| Location                                                                                                             |        |         | Sandwich, UK                   |                                       |                           |             |                            |                    |                                                                                                                                                                                                                                                                                                                                                       |                 |                     |          |
| Project Name                                                                                                         |        |         | PDF Extension Facility         |                                       |                           |             |                            |                    |                                                                                                                                                                                                                                                                                                                                                       |                 |                     |          |
| Project No.                                                                                                          |        |         | 300291                         |                                       | A1                        | 21 May 26   | Feasibility Design Issue   | ME                 | MC                                                                                                                                                                                                                                                                                                                                                    | ME              | DOCUMENT NUMBER     |          |
|                                                                                                                      |        |         |                                |                                       |                           |             |                            |                    |                                                                                                                                                                                                                                                                                                                                                       |                 | 300291-PR-SC-0001   |          |
|                                                                                                                      |        |         |                                |                                       |                           |             |                            |                    |                                                                                                                                                                                                                                                                                                                                                       |                 | A1                  |          |
| Tag                                                                                                                  | ID No. | No. Off | DESCRIPTION                    | TECHNICAL CHARACTERISTICS - SIZE/DUTY | MATERIALS OF CONSTRUCTION | WEIGHT (kg) | DIMENSIONS (mm) W x D x H  | LOCATION           | Process                                                                                                                                                                                                                                                                                                                                               | Electrical Load | NOTES               | REVISION |
|                                                                                                                      | P1     | 1       | Effluent Transfer Pump         | 20m³/h @ 19.8m TDH<br>Self-priming    | Hastelloy-C22             |             |                            | Externally located |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |
|                                                                                                                      | P2     | 1       | Effluent Transfer Pump         | 20m³/h @ 19.8m TDH<br>Self-priming    | Hastelloy-C22             |             |                            | Externally located |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |
| TT04                                                                                                                 |        | 1       | Trade Effluent Test Tank       | 35m³ working volume                   | Glass-Lined Carbon Steel  |             | 3000 D x 5700 H (internal) | Externally located |                                                                                                                                                                                                                                                                                                                                                       |                 | For extension alone | A1       |
| TT05                                                                                                                 |        | 1       | Trade Effluent Test Tank       | 35m³ working volume                   | Glass-Lined Carbon Steel  |             | 3000 D x 5700 H (internal) | Externally located |                                                                                                                                                                                                                                                                                                                                                       |                 | For extension alone | A1       |
| TT04                                                                                                                 | P1A    | 1       | Effluent Transfer Pump         | 60m³/h @ 36.4m TDH                    | Hastelloy-C22             |             |                            | Externally located |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |
| TT04                                                                                                                 | P1B    | 1       | Effluent Transfer Pump         | 60m³/h @ 36.4m TDH                    | Hastelloy-C22             |             |                            | Externally located |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |
| SP02                                                                                                                 |        | 1       | Sump Tank - Effluent Tank Sump |                                       | SS316L                    |             |                            | Externally located |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |
| CONTAINMENT & ISOLATION                                                                                              |        |         |                                |                                       |                           |             |                            |                    |                                                                                                                                                                                                                                                                                                                                                       |                 |                     |          |
| -                                                                                                                    |        | 1       | Solid packing glove box        |                                       | SS316L                    |             |                            | Ground floor       |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |
| -                                                                                                                    |        | 1       | Solid packing glove box        |                                       | SS316L                    |             |                            | Ground floor       |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |
| -                                                                                                                    |        | 1       | Crushing packaging isolator    |                                       | SS316L                    |             |                            | Ground floor       |                                                                                                                                                                                                                                                                                                                                                       |                 |                     | A1       |

Notes:  
(1) The selections for equipment such as the condenser and tanks are currently preliminary estimates and will require detailed calculations and assessment in the next phase.  
(2) The equipment numbering is provisional/assumed.